



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2025

3rd Year 1st Semester, Session: 2021-2022

Course Code: CSE-3103

Time: 3.00 hours

Course Title: Computer Networks

Marks: 72

[N.B. Answer any 6 set questions from following questions. The figures in the margin indicate full marks.]

1.

- a) Write True or False for the following statements: 06
- In packet-switched networks, the total end-to-end delay is solely determined by the transmission delay and does not include propagation or queuing delays.
 - HTTP/3 uses TCP as its underlying transport protocol to ensure reliable data delivery.
 - In the Selective Repeat (SR) protocol, the receiver discards all out-of-order packets.
 - The IP protocol guarantees the delivery of datagrams to their destination without loss or duplication.
 - In Ethernet networks, collisions can occur only in half-duplex mode, not in full-duplex mode.
 - WPA3 provides stronger security than WPA2 by using individualized data encryption for each user on a WiFi network.
- b) Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20 percent of the time. 04
- When circuit switching is used, how many users can be supported?
 - For the remainder of this problem, suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time? Why will there be a queuing delay if three users transmit at the same time?
 - Find the probability that a given user is transmitting.
 - Suppose now there are three users. Find the probability that at any given time, all three users are transmitting simultaneously. Find the fraction of time during which the queue grows.

- c) Consider a Car-Caravan analogy where the propagation speed is 100 km/hour. Suppose the caravan travels 175 km, beginning in front of one tollbooth, passing through a second tollbooth, and finishing just after a third tollbooth. What is the end-to-end delay? 02

2.

- a) Suppose Alice, with a Web-based e-mail account (such as Hotmail or Gmail), sends a message to Bob, who accesses his mail from his mail server using IMAP. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts. 04
- b) What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it? 03
- c) Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B. 05
- Express the propagation delay, d_{prop} , in terms of m and s .
 - Determine the transmission time of the packet, d_{trans} , in terms of L and R .
 - Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
 - Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{trans}$, where is the last bit of the packet?
 - Suppose d_{prop} is greater than d_{trans} . At time $t = d_{trans}$, where is the first bit of the packet?

3.

- a) Suppose, you are provided a private network 192.168.10.0. Subnet the following network using: 08
- Classful Addressing.
 - Classless Addressing.



4.

- b) Explain IEEE802.3 Ethernet Frame format. 04
- a) Suppose Client A requests a web page from Server S through HTTP and its socket is associated with port 33000. 04
- What are the source and destination ports for the segments sent from A to S?
 - What are the source and destination ports for the segments sent from S to A?
 - Can Client A contact to Server S using UDP as the transport protocol?
 - Can Client A request multiple resources in a single TCP connection?
- b) Assume that a host receives a UDP segment with 01011101 and 11110010 as the checksum. 03
- The host adds the 16-bit words over all necessary fields excluding the checksum and obtains the value 00110010 and 00001101. Is the segment considered correctly received or not? What does the receiver do?

- c) Consider a stop-and-wait data-transfer protocol that provides error checking and retransmissions but uses only negative acknowledgments. Assume that negative acknowledgments are never corrupted. Would such a protocol work over a channel with bit errors? What about over a lossy channel with bit errors? 03
- d) Consider a scenario where the three hosts A, B, and C are connected as a ring: A to B, B to C, and C to A. Assume that A and C run protocol rdt3.0, whereas B simply relays all messages received from A to C. 02
- Does this arrangement enable reliable delivery of messages from A to C?
 - Can B tell if a certain message has been correctly received by A?

5. a) What is the role of the forwarding table within a router? 02
- b) What is meant by the term "route aggregation"? Why is it useful for a router to perform route aggregation? 03
- c) Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3, and packets are to be forwarded to the link interfaces as follows: 05

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

- Provide a forwarding table that has five entries, uses longest prefix matching, and forwards packets to the correct link interfaces.
 - Describe how your forwarding table determines the appropriate link interface for datagrams with destination addresses:
11001000 10010001 01010001 01010101
11100001 01000000 11000011 00111100
11100001 10000000 00010001 01110111
- d) Suppose two packets arrive to two different input ports of a router at exactly the same time. Also suppose there are no other packets anywhere in the router. 02
- Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a shared bus?
 - Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses switching via memory?
 - Suppose the two packets are to be forwarded to the same output port. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a crossbar?

6. a) Switching in a router forwards data from an input port to an output port. What is the advantage of switching via an interconnection network over switching via memory and switching via bus? 02

- b) Compare and contrast the IPv4 and the IPv6 header fields. Do they have any fields in common? 03

- c) Suppose that the WEQ scheduling policy is applied to a buffer that supports three classes, and suppose the weights are 0.5, 0.25, and 0.25 for the three classes. 04

- Suppose that each class has a large number of packets in the buffer. In what sequence might the three classes be served in order to achieve the WFQ weights? (For round robin scheduling, a natural sequence is 123123123 . . .).
- Suppose that classes 1 and 2 have a large number of packets in the buffer, and there are no class 3 packets in the buffer. In what sequence might the three classes be served in to achieve the WFQ weights?

- d) What is the role of a *packet scheduler* at the output port of a router? 03

7. a) Compare UDP and TCP in terms of their features and use cases. Explain how TCP ensures reliable data transfer, while UDP doesn't. 04

- b) Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 4) and Go-Back N error control strategy. All packets are ready and immediately available for transmission. If every 3rd packet that A transmits gets lost (but no ACKs from B ever get lost), then calculate the number of packets that A will transmit for sending the message to B. 08

8. a) What is DES? Explain why 3-DES is secured than DES? 03

- b) Explain different types of Authentication Protocol in network security. Why AP-5 is more secure than AP-4, AP-3, AP-2, and AP-1? 03

- c) Man in the Middle attack can be seen in AP-5. Explain how to deal with the attack using digital signatures and sign message digest. 03

- d) i) Using RSA, choose $p = 5$ and $q = 7$, and encode the numbers 12, 19, and 27 separately. Apply the decryption algorithm to the encrypted version to recover the original plaintext message. 03

- ii) Choose p and q of your own and encrypt 1834 as one message m .